

Anselm Paulus

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✂ [AnselmPaulus](#) • [a-paulus](https://github.com/a-paulus) • Last updated on May 26, 2026

Current Position

Ph.D. Student in Machine Learning, *University of Tuebingen & Max-Planck-Institute for Intelligent Systems*, Tuebingen, Germany (supervised by [Georg Martius](#) on differentiable optimization and LLM safety/security) 2022 – 2026

Education

M.Sc. in Computer Science, *University of Tuebingen* (4.00/4.00) 2021 – 2022

B.S. in Computer Science, *University of Tuebingen* (3.73/4.00) 2018 – 2021

B.S. in Physics, *University of Tuebingen* (3.80/4.00) 2015 – 2021

Previous Positions

Research Scientist Intern, *Meta, Fundamental AI Research (FAIR)*, Menlo Park, CA (with [Arman Zharmagambetov](#) on game-theoretical perspectives on LLM safety) 2025 – 2026

Research Scientist Intern, *Meta, Fundamental AI Research (FAIR)*, New York City (with [Brandon Amos](#) on amortized optimization for LLM safety) 2023 – 2024

Research Assistant, *Max-Planck-Institute for Intelligent Systems*, Tuebingen, Germany (with [Georg Martius](#) on differentiable combinatorial optimization) 2018 – 2022

Publications [[Google Scholar](#): 1k+ citations]

Selected publications I am a primary author on are highlighted.

2026.....

1. *Hard Contacts with Soft Gradients: Refining Differentiable Simulators for Learning and Control*
Anselm Paulus*, [Andreas René Geist*](#), [Pierre Schumacher](#), [Vít Musil](#), and [Georg Martius](#)
ICLR 2026
2. *Safety Alignment of LMs via Non-cooperative Games*
Anselm Paulus, [Ilia Kulikov](#), [Brandon Amos](#), [Rémi Munos](#), [Ivan Evtimov](#), [Kamalika Chaudhuri](#), and [Arman Zharmagambetov](#)
ICML 2026 (Spotlight)

3. *SoftJAX & SoftTorch: Empowering Automatic Differentiation Libraries with Informative Gradients*
Anselm Paulus, [A. René Geist](#), [Vít Musil](#), [Sebastian Hoffmann](#), [Onur Beker](#), and [Georg Martius](#)
ICML 2026 (Oral)

2025.....

4. *AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs* [code]
Anselm Paulus*, [Arman Zharmagambetov*](#), [Chuan Guo](#), [Brandon Amos†](#), and [Yuandong Tian†](#)
ICML 2025

- 2024.....
5. [LPGD: A General Framework for Backpropagation through Embedded Optimization Layers](#) [code]
Anselm Paulus, Georg Martius, and Vít Musil
ICML 2024
- 2023.....
6. [Backpropagation through Combinatorial Algorithms: Identity with Projection Works](#) [code]
Anselm Paulus*, Subham Sekhar Sahoo*, Marin Vlastelica, Vít Musil, Volodymyr Kuleshov, and Georg Martius
ICLR 2023
- 2021.....
7. [CombOptNet: Fit the Right NP-Hard Problem by Learning Integer Programming Constraints](#) [code]
Anselm Paulus, Michal Rolínek, Vít Musil, Brandon Amos, and Georg Martius
ICML 2021 (Spotlight, Oral at LMCA NeurIPS 2020 workshop)
- 2020.....
8. [Differentiation of Blackbox Combinatorial Solvers](#) [code]
Anselm Paulus*, Marin Vlastelica P.*, Vít Musil, Georg Martius, and Michal Rolínek
ICLR 2020 (Spotlight)
 9. [Optimizing Rank-based Metrics with Blackbox Differentiation](#) [code]
Michal Rolínek*, Vít Musil*, **Anselm Paulus**, Marin Vlastelica P., Claudio Michaelis, and Georg Martius
CVPR 2020 (Oral, best paper award nomination)
 10. [Deep Graph Matching via Blackbox Differentiation of Combinatorial Solvers](#) [code]
Michal Rolínek, Paul Swoboda, Dominik Zietlow, **Anselm Paulus**, Vít Musil, and Georg Martius
ECCV 2020

Open Source Repositories

0.7k+ GitHub stars across all repositories.

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|---|------|
| 1. a-paulus/softjax — ★32 — <i>SoftJAX (library for differentiable operations in JAX)</i> | 2026 |
| 2. a-paulus/softtorch — ★28 — <i>SoftTorch (library for differentiable operations in PyTorch)</i> | 2026 |
| 3. facebookresearch/advgame — ★17 — <i>AdvGame (Adversarial attacks on LLMs)</i> | 2026 |
| 4. facebookresearch/advprompter — ★180 — <i>AdvPrompter (Adversarial attacks on LLMs)</i> | 2025 |
| 5. martius-lab/diffcp-lpgd — ★2 — <i>Lagrangian Proximal Gradient Descent (now merged into CVXPY)</i> | 2024 |
| 6. martius-lab/solver-differentiation-identity — ★11 — <i>Blackbox Identity Differentiation</i> | 2023 |
| 7. martius-lab/CombOptNet — ★74 — <i>CombOptNet</i> | 2021 |
| 8. martius-lab/blackbox-deep-graph-matching — ★88 — <i>Deep Graph Matching</i> | 2020 |
| 9. martius-lab/blackbox-backprop — ★349 — <i>Blackbox Differentiation</i> | 2020 |

Invited Talks

1. *AdvPrompter: Fast Adaptive Adversarial Prompting for LLMs* — Masaryk University, Brno, Czechia 2024
2. *On differentiable combinatorial optimization* — Dagstuhl Seminar: Machine Learning and Logical Reasoning: The New Frontier 2022

Professional Activities

Reviewing.....

International Conference on Learning Representations (ICLR): 2022, 2023, 2024, 2025

International Conference on Machine Learning (ICML): 2022, 2023, 2024, 2025

Neural Information Processing Systems (NeurIPS): 2021, 2022, 2025

Neural Information Processing Systems (NeurIPS) DiffCoAlg Workshop: 2025

Skills

Programming C++, Java, Python

Frameworks JAX, NumPy, Pandas, PyTorch, SciPy, TensorFlow

Toolbox Linux, emacs, git, tmux, zsh, uv, vscode